printed page 1104 equation (5.5)

$$\rho_{X_0,t_0}(F,t) = \frac{1}{4\pi(t_0-t)} \exp\left(-\frac{|F-X_0|^2}{4(t_0-t)}\right),$$

cutoff $\phi \in C_0^\infty(B_{2r}(X_0))$ $\phi \equiv 1$ on $B_r(X_0)$ localized Gaussian quantity

$$\Psi(X_0,t_0;t) = \int_{\Sigma_t} (\cos\alpha)^{-p}\phi(F)\rho_{X_0,t_0}(F,t)\,d\mu_t.$$

Theorem 3.3 differential inequality

$$\frac{d}{dt}\left(e^{c_1\sqrt{t_0-t}}\Psi(X_0,t_0;t)\right) \le -\text{nonnegative square terms} + c_2 e^{c_1\sqrt{t_0-t}}.$$

Section 5 parabolic rescaling

$$F_\lambda(x,s) = \lambda\big(F(x,T+\lambda^{-2}s)-X_0\big), \qquad s<0,$$
$$d\mu_s^\lambda = \lambda^2 d\mu_{T+\lambda^{-2}s}, \qquad w_\lambda = (\cos\alpha_\lambda)^{-p}.$$

printed equation (5.5) fixed $-1 < s_1 < s_2 < 0$

$$G_\lambda(s_2) - G_\lambda(s_1) \to 0 \quad \text{as } \lambda \to \infty,$$

endpoint quantity

$$G_\lambda(s) = e^{c_1\sqrt{T-(T+\lambda^{-2}s)}} \int_{\Sigma_s^\lambda} w_\lambda\,\phi_R(F_\lambda)\,(-s)^{-1}\exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right)d\mu_s^\lambda.$$

gradient flow printed page 1104 equation (5.5)

exact gap corrected claim remaining moving-cutoff lemma

lemma false / underived strongest corrected theorem

Proven **cutoff**

$\chi \in C_0^\infty(B_{2r}(X_0))$ admissible cutoff $\chi = 1$ on $B_r(X_0)$

$$\Psi_\chi(t) = \int_{\Sigma_t} (\cos\alpha)^{-p}\chi(F)\frac{1}{4\pi(T-t)}\exp\left(-\frac{|F-X_0|^2}{4(T-t)}\right)d\mu_t,$$

$$Q_\chi(t) = e^{c_1\sqrt{T-t}}\Psi_\chi(t).$$

differential inequality nonpositive square terms

$$Q_\chi'(t) \le c_2 e^{c_1\sqrt{T-t}}.$$

$[a,T)$ integrable

$$A(t) = \int_t^T c_2 e^{c_1\sqrt{T-u}}\,du.$$

$A(t) \to 0$

$$(Q_\chi + A)'(t) = Q_\chi'(t) - c_2 e^{c_1\sqrt{T-t}} \le 0.$$

$Q_\chi + A$ monotone decreasing bounded below $Q_\chi(t)$ finite one-sided limit

$$\lim_{t\uparrow T} Q_\chi(t) = L_\chi < \infty.$$

exact rescaling

$$t_\lambda(s) = T + \lambda^{-2}s.$$

$$T - t_\lambda(s) = \lambda^{-2}(-s), \qquad F - X_0 = \lambda^{-1}F_\lambda, \qquad d\mu_s^\lambda = \lambda^2 d\mu_{t_\lambda(s)}.$$

Kähler angle positive homothety invariant

$$(\cos\alpha_\lambda)^{-p} = (\cos\alpha)^{-p}.$$

$$\int_{\Sigma_s^\lambda} (\cos\alpha_\lambda)^{-p}\chi(X_0 + \lambda^{-1}F_\lambda)(-s)^{-1}\exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right) d\mu_s^\lambda$$
$$= 4\pi\,\Psi_\chi(t_\lambda(s)).$$

exponential factor

$$J_\lambda^\chi(s) = 4\pi Q_\chi(t_\lambda(s)).$$

fixed $s_1 < s_2 < 0$

$$J_\lambda^\chi(s_2) - J_\lambda^\chi(s_1) = 4\pi\big(Q_\chi(t_\lambda(s_2)) - Q_\chi(t_\lambda(s_1))\big) \to 0,$$

terminal limit L_χ

4π normalization normalized kernel $[4\pi(-s)]^{-1}$ factor unnormalized $(-s)^{-1}$ 4π zero limit

Proven printed fixed scaled cutoff fixed original cutoff

printed expression $\phi_R(F_\lambda)$ ϕ_R fixed in scaled coordinates original coordinates

$$\phi_R(F_\lambda) = \phi_R(\lambda(F - X_0)) =: \chi_{R,\lambda}(F).$$

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

cutoff $B_{2R/\lambda}(X_0)$ $B_{R/\lambda}(X_0)$ 1 λ

printed endpoint quantity

$$4\pi$$

Section 3 type quantity quantity cutoff $\chi_{R,\lambda}$ fixed χ $Q_\chi(t)$ terminal-limit argument moving family $Q_{\chi_{R,\lambda}}(t)$

Proven individual terminal limits for moving families two-time limit

$$Q_n(t) = n^2(-t), \qquad t \in (-1, 0).$$

$Q_n \geq 0$ locally absolutely continuous

$$Q_n'(t) = -n^2 \leq 0.$$

Q_n terminal limit

$$\lim_{t \uparrow 0} Q_n(t) = 0.$$

fixed $s_1 < s_2 < 0$

$$Q_n(n^{-2}s_2) - Q_n(n^{-2}s_1) = s_1 - s_2 \neq 0.$$

“ moving cutoff quantity terminal limit” imply diagonal parabolic-scale two-time convergence

Proven literal fixed scaled compact cutoff static plane model

flat symplectic plane $P = \mathbb{R}^2 \times \{0\} \subset \mathbb{C}^2$ stationary immersion $F(x,t) = x$

$$\cos\alpha = 1, \qquad H = 0, \qquad V = 0,$$

parabolic rescaling plane weight 1

nontrivial radial compact cutoff ϕ_R

$$\phi_R = 1 \text{ on } B_R, \qquad \phi_R = 0 \text{ outside } B_{2R},$$

radial nonincreasing

$$I(s) = \int_P \phi_R(Y)(-s)^{-1} \exp\left(-\frac{|Y|^2}{4(-s)}\right) dA(Y).$$

uncut integral

$$\int_P (-s)^{-1} \exp\left(-\frac{|Y|^2}{4(-s)}\right) dA(Y) = 4\pi.$$

compactly cut integral $s_1 = -1$ $s_2 = -1/4$ $Y = \sqrt{a}Z$ $a = -s$

$$I(-a) = \int_P \phi_R(\sqrt{a}\, Z) e^{-|Z|^2/4} \, dA(Z).$$

$a_2 = 1/4 < a_1 = 1$ ϕ_R radial nonincreasing

$$\phi_R(\sqrt{a_2}Z) \geq \phi_R(\sqrt{a_1}Z),$$

cutoff annulus

$$I(-1/4) > I(-1).$$

static plane model

$$G_\lambda(s) = e^{c_1 \lambda^{-1}\sqrt{-s}} I(s),$$

$$\lim_{\lambda\to\infty} \big(G_\lambda(-1/4) - G_\lambda(-1)\big) = I(-1/4) - I(-1) > 0.$$

disproves standalone fixed-scaled-cutoff lemma

plane full Section 5 compact first-singular-flow counterexample noncompact finite first singular time X_0 T singular point

Proven moving cutoff derivative terms order-one annular flux

literal cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)),$$

$$D\chi_{R,\lambda} = \lambda D\phi_R, \qquad D^2\chi_{R,\lambda} = \lambda^2 D^2\phi_R.$$

pre-(5.5) monotonicity calculation cutoff-gradient cutoff-Laplacian heat-kernel-gradient terms parabolic change of variables $t = T + \lambda^{-2}s$ fixed scaled annulus

$$R < |F_\lambda| < 2R$$

spacetime integrals

$$\int_{s_1}^{s_2} \int_{\Sigma_s^\lambda} w_\lambda \rho_s(F_\lambda) \left[D\phi_R(F_\lambda)\cdot v_\lambda + \Delta_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) + 2\langle \nabla_{\Sigma_s^\lambda} \log \rho_s, \nabla_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda)\rangle \right] d\mu_s^\lambda ds.$$

λ^{-1} λ^{-2} prefactor moving cutoff error scaling $o(1)$ literal moving-cutoff endpoint limit annular flux vanishing / cancellation uniform moving-cutoff convergence-to-terminal-limit

Conditional endpoint weighted measures nonzero cone/plane literal compact cutoff positive defect

s_1, s_2 endpoint pushed-forward weighted measures nonzero two-dimensional cone measure ν_{cone} suitable radial compact cutoff

$$\int \phi_R(Y)(-s_2)^{-1}e^{-|Y|^2/[4(-s_2)]}\,d\nu_{\text{cone}} > \int \phi_R(Y)(-s_1)^{-1}e^{-|Y|^2/[4(-s_1)]}\,d\nu_{\text{cone}}.$$

literal compact scaled cutoff endpoint difference limit strictly positive zero conditional obstruction full counterexample same-limit conical convergence (5.5) noncircularly

printed $\phi_R(F_\lambda)$ fixed in scaled coordinates original coordinates moving cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

Theorem 3.3 terminal-limit argument fixed localized Gaussian quantity

$$Q_\chi(t) \to L_\chi,$$

moving family $Q_{\chi_{R,\lambda}}(t)$ parabolic-scale times uniformly terminal limit

moving cutoff monotonicity calculation cutoff derivative terms scaled variables order-one annular flux small parameter sign static plane computation flux velocity curvature flow-energy compact cutoff Gaussian width s endpoint difference

fixed-cutoff monotonicity one-sided limit local area bound fixed-time Radon compactness literal (5.5) noncircular bridge

stage	question addressed	conclusion established	effect on the approach
fixed-cutoff terminal limit	(3.6) one-sided inequality terminal limit	fixed cutoff χ $Q_\chi(t)$ has finite one-sided limit as $t \uparrow T$	fixed-cutoff endpoint route is valid
exact rescaling	rescaled endpoint quantity Section 3 quantity	unnormalized scaled integral equals $4\pi\Psi_\chi$ for fixed pullback cutoff $\chi(X_0 + \lambda^{-1}F_\lambda)$	4π is harmless normalization

stage	question addressed	conclusion established	effect on the approach
cutoff comparison	printed $\phi_R(F_\lambda)$ fixed-cutoff pullback	$\phi_R(F_\lambda)$ corresponds to moving cutoff $\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$	identifies the real gap
formal moving-family test	individual terminal limits imply diagonal two-time convergence	false, even for monotone nonnegative scalar families	fixed-cutoff argument cannot be applied familywise
static plane model	fixed scaled compact cutoff intrinsically harmless	false in standalone model; endpoint difference can be strictly positive	shows the obstruction is real
annular flux audit	pre-(5.5) estimates hide a uniform moving-cutoff estimate	no; derivative terms become order-one annular flux	literal moving-cutoff lemma remains unproved
compactness audit	fixed-time local area compactness repairs the gap	only gives separate fixed-time weak limits, not same-limit or flux control	insufficient for literal endpoint limit
corrected theorem	what unconditional statement is actually proved	fixed-original-cutoff endpoint theorem with $\chi(X_0 + \lambda^{-1}F_\lambda)$	valid replacement for rigorous use

literal fixed compact scaled cutoff reading verified proof full compact first-singular-flow counterexample proved stronger diagnosis (3.6) printed (5.5) common-limit step fixed-original-cutoff replacement literal $\phi_R(F_\lambda)$ noncircular moving-cutoff theorem

unconditional corrected statement \boxed{ \begin{minipage}{0.92\linewidth} Let \(X_0\) be the blow-up point and \(T\) the first singular time. Let \(\chi\in C_0^\infty(B_{2r}(X_0))\) be fixed with \(\chi=1\) on \(B_r(X_0)\). For fixed \(-1<s_1<s_2<0\), define @@MATH0@@ Then @@MATH1@@ The same statement with normalized kernel \([4\pi(-s)]^{-1}\) is equivalent. \end{minipage} }

literal moving-cutoff lemma \boxed{ \begin{minipage}{0.92\linewidth} For the moving cutoff @@MATH2@@ let @@MATH3@@ If \(L_{R,\lambda}=\lim_{t\uparrow T}Q_{R,\lambda}(t)\), prove noncircularly that for each \(i=1,2\), @@MATH4@@ Equivalently, prove annular cutoff-flux cancellation/vanishing on @@MATH5@@ at the parabolic scale \(T-t\sim\lambda^{-2}\). \end{minipage} } bridge proved full-hypothesis counterexample constructed printed literal compact-scaled-cutoff equation (5.5)